

Lesson 2 Docker 'hello-world'

The "hello-world" provided by Docker is a lightweight container, which is designed to test the successful installation and operation of Docker. It is very small in size and primarily used to demonstrate Docker's basic usage and verify the correctness of the Docker operating environment.

1) Press "Ctrl+Alt+T" to open the command line terminal. Then enter "docker run hello-world" to download the "hello-world" image (if it is not present on your local machine).

```
pi@raspberrypi:~ $ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (arm64v8)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

2) Enter "docker images" in the terminal to list all images on the local master.

```
pi@raspberrypi:~ $ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
hello-world         latest         ee301c921b8a   8 months ago   9.14kB
```

The relevant information of "Hello_World" can be checked, corresponding to the parameters:

REPOSITORY: image's repository source.

TAG: image's tag.

IMAGE ID: image's ID.

CREATED: image's creation time.

SIZE: image's size.

3) Enter “docker run -it hello-world” in the terminal to run the “hello-world” image. The terminal will print relevant prompt information.

```
pi@raspberrypi:~ $ docker run -it hello-world

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```

4) Enter “docker inspect ee30efe9” in the terminal to output JSON data including detailed information about the “hello-world” image.

```
pi@raspberrypi:~ $ docker inspect ee301c921b8a
[
  {
    "Id": "sha256:ee301c921b8aad002973b2e0c3da17d701dcd994b606769a7e6eaa100b81d44",
    "RepoTags": [
      "hello-world:latest"
    ],
    "RepoDigests": [
      "hello-world@sha256:4bd78111b6914a99dbc560e6a20eab57ff6655aea4a80c50b0c5491968cbc2e6"
    ],
    "Parent": "",
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2023-05-02T16:49:27Z",
    "Container": "",
    "ContainerConfig": {
      "Hostname": "",
      "Domainname": "",
      "User": "",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "Tty": false,
      "OpenStdin": false,
      "StdinOnce": false,
      "Env": null,
      "Cmd": null,
      "Image": "",
      "Volumes": null,
      "WorkingDir": "",
      "Entrypoint": null
    }
  }
]
```